ScamScan: AI-Powered Scam Detection

Detect scam messages, emails, and URLs instantly with ScamScan, presented by Scamalytics at the Techbyte hackathon.

By Sathwika Reddy

The Growing Threat of Online Scams

Challenge

Users face daily scam messages and phishing emails.

Statistics

- 75% of phishing attacks start with emails
- Billions lost annually to scams

Need

A fast, reliable scam detection tool is crucial.

ScamScan: Instant Scam Detection

ML-Powered

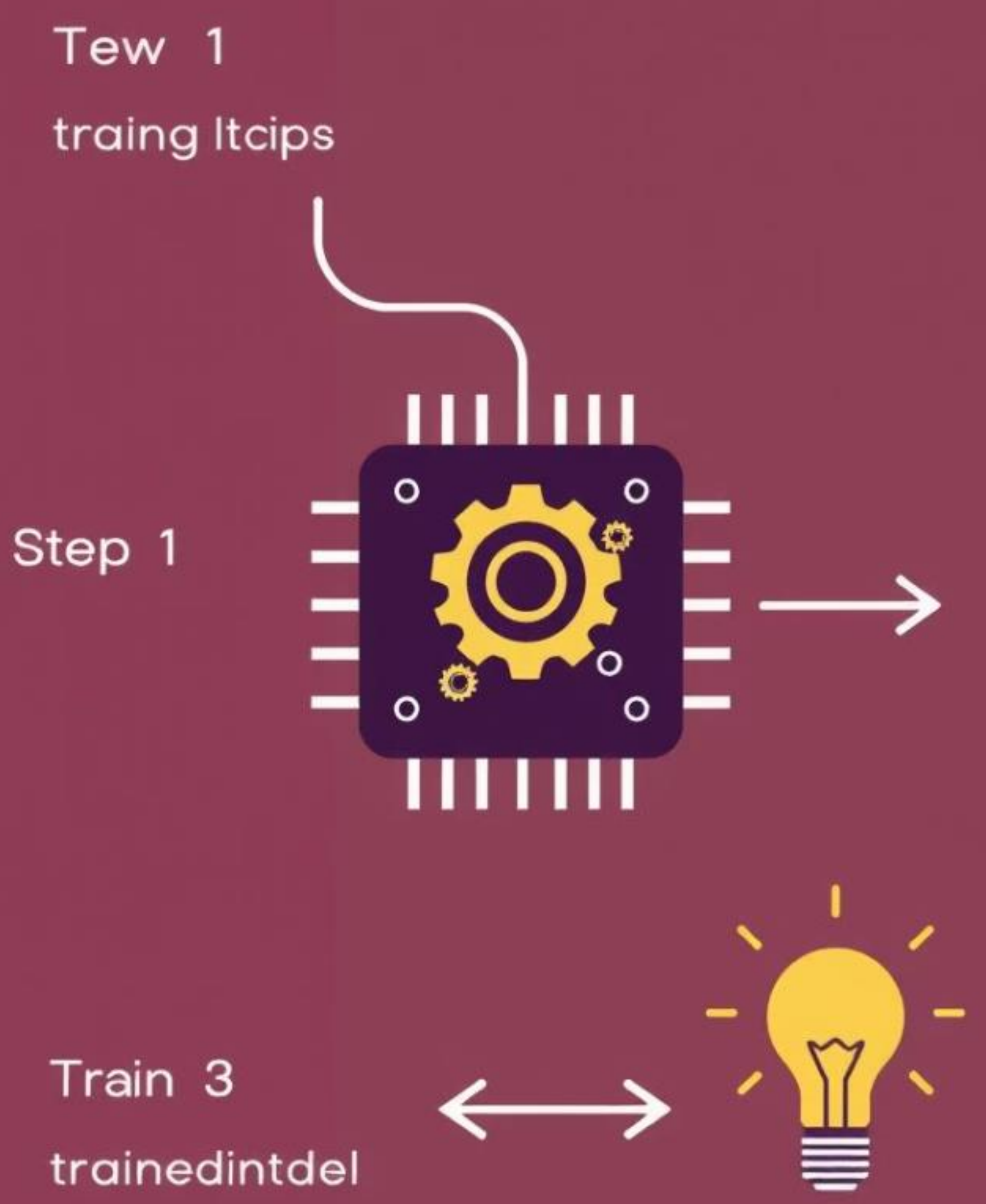
Detect scam content instantly with machine learning.

Versatile

Works with messages, emails, and bulk uploads.

User-Friendly

Simple, intuitive interface for all users.



How ScamScan Works

1

Input

User pastes message or email.

2

Preprocessing

Text cleaned and vectorized.

3

Model

Trained ML model (Logistic Regression).

4

Output

Scam or Not Scam with confidence score.

Our Tech Stack

Streamlit

Frontend (Python)

Flask

Backend API

Scikit-learn

ML Library

Render

Deployment

Demo Screens

Streamlit UI

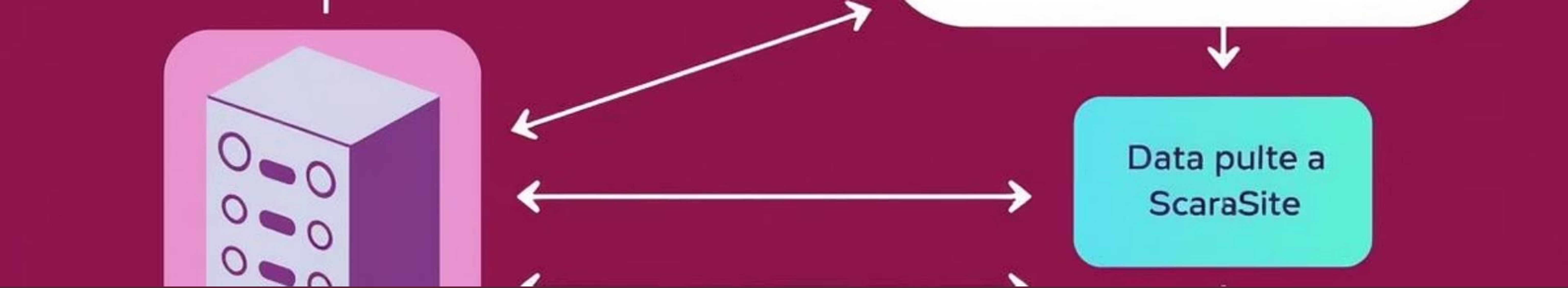
User-friendly input interface.

Flask API Response

JSON output for analysis.

Jupyter Notebook

Model training visualization.



Deployment Details

Backend

Deployed on Render for API access.

Frontend

Hosted on Streamlit Cloud for public testing.

Accessibility

Publicly accessible for testing and feedback.

Challenges Overcome

Model Integration

Successfully integrated ML model with Flask.

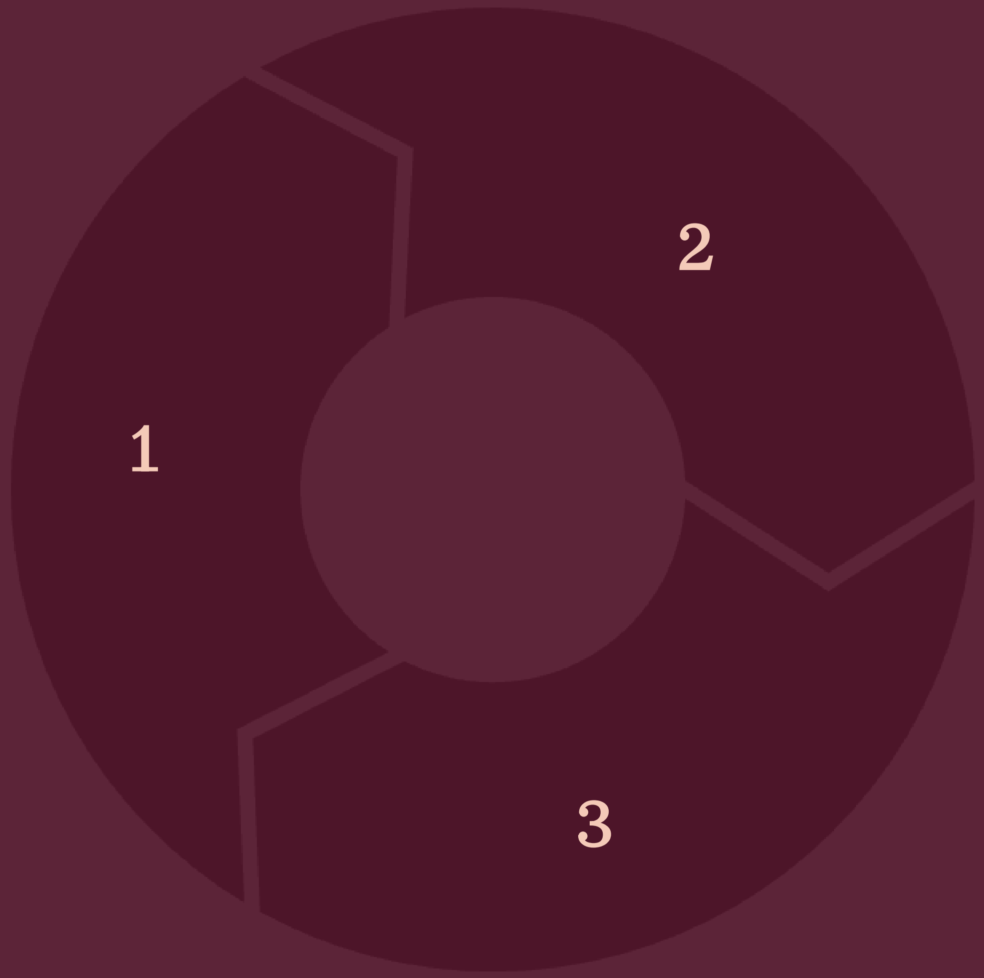
Frontend Sync

Achieved seamless frontend-backend synchronization.

Online Hosting

Deployed ML pipeline online efficiently.

Impact and Use Cases



Personal Use

Scan SMS, emails for scams.

Enterprise

Protect banks, customer support.

Expansion

Job scams, phishing URLs.

Next Steps and Future Vision

1 Enhance Model

Improve detection accuracy.

2 Expand Coverage

Add more scam types.

3 Partnerships

Collaborate with institutions.